

Proposed internship topic for master's 2/engineering school 4 to 6 months starting in early 2027

Project title: Leveraging External Data in Clinical Trials: Application of Causal Inference Methods in the Bayesian Framework to Improve Statistical Efficiency

Team : Institut du Cerveau/INRIA – Équipe ARAMIS

Team leader : Sophie Tezenas du Montcel

Supervisor : Maylis Tran

Internship objectives :

Randomized controlled trials (RCTs) are widely regarded as the gold standard for evaluating treatment effects, as randomization helps minimize bias and provides a strong basis for causal inference. However, the design and conduct of RCTs can raise important ethical and practical challenges. In particular, placebo-controlled trials may be ethically difficult when promising new treatments are available. These challenges are especially pronounced in rare neurodegenerative diseases, where the limited number of eligible patients can make patient recruitment difficult and substantially constrain the statistical power of clinical trials. Improving the statistical efficiency of clinical trials is therefore an important methodological challenge. A variety of statistical approaches have been developed to increase precision and make better use of the information available to researchers. In this context, the use of external data has received increasing attention.

Several types of external data can be incorporated into clinical trials. For example, external control data can be used to construct hybrid or externally controlled trials [1,2]. More recently, the growing availability of observational and longitudinal data, together with advances in machine learning, has opened up new opportunities. These data can provide detailed information on disease progression and can be used to develop predictive models of patients' future disease states. Such models can be incorporated into clinical trials to compare an observed treatment response with the outcome that would be expected in the absence of treatment [3,4]. A growing methodological literature has investigated estimators that integrate external information into clinical trials, using both frequentist and Bayesian approaches [5]. However, important methodological questions remain regarding the conditions under which these approaches can improve statistical efficiency while maintaining adequate control of bias and type I error [6].

The objective of this internship is to review and investigate existing Bayesian methods for incorporating external information into clinical trials, with a particular focus on approaches based on causal inference. The different methodologies will first be studied and compared theoretically, then through simulation studies and real-world data from patients with ataxia. Finally, the results will be compared with frequentist approaches that have already been studied within the laboratory.

This internship will therefore provide an opportunity to work at the intersection of **causal inference, Bayesian statistics, clinical trials, and machine learning**, with a particular focus on developing statistically efficient methods for clinical research in rare diseases.

References

- [1] Concept paper on the development of a reflection paper on the use of External controls for evidence generation in regulatory decision-making. Reference Number: EMA/125200/2026.
- [2] FDA guidance document - Considerations for the Design and Conduct of Externally Controlled Trials for Drug and Biological Products, 2023.
- [3] Poulet, PE., Tran, M., Tezenas du Montcel, S. et al. Prediction-powered inference for clinical trials: application to linear covariate adjustment. BMC Med Res Methodol 25, 204 (2025). <https://doi.org/10.1186/s12874-025-02647-6>
- [4] Schuler A, Walsh D, Hall D, Walsh J, Fisher C; Critical Path for Alzheimer's Disease; Alzheimer's Disease Neuroimaging Initiative; Alzheimer's Disease Cooperative Study. Increasing the efficiency of randomized trial estimates via linear adjustment for a prognostic score. Int J Biostat. 2021 Dec 22;18(2):329-356. doi: 10.1515/ijb-2021-0072. PMID: 34957728.
- [5] Bénédicte Colnet. Imke Mayer. Guanhua Chen. Awa Dieng. Ruohong Li. Gaël Varoquaux. Jean-Philippe Vert. Julie Josse. Shu Yang. "Causal Inference Methods for Combining Randomized Trials and Observational Studies: A Review." Statist. Sci. 39 (1) 165 - 191, February 2024. <https://doi.org/10.1214/23-STS889>
- [6] Kopp-Schneider A, Calderazzo S, Wiesenfarth M. Power gains by using external information in clinical trials are typically not possible when requiring strict type I error control. Biom J. 2020 Mar;62(2):361-374. doi: 10.1002/bimj.201800395. Epub 2019 Jul 2. PMID: 31265159; PMCID: PMC7079072.

Prerequisites : Python, R, Causal Inference, Bayesian statistics

Contact : Maylis Tran maylis.tran@icm-institute.org